

LM12 Introduction to Financial Statement Modeling

1. Introduction	2
2. Building a Financial Statement Model	2
3. Behavioral Finance and Analyst Forecasts	8
4. The Impact of Competitive Factors in Prices and Costs	9
5. Modeling Inflation and Deflation.....	9
6. The Forecast Horizon and Long-Term Forecasting	11
Summary	13

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1. Introduction

The learning module introduces us to financial modeling; we will learn how to forecast each item of the income statement, balance sheet, and cash flow statement, and subsequently construct a pro forma income statement, balance sheet and cash flow statement.

Then, we will discuss five key behavioral biases that influence the modeling process and strategies to mitigate them.

We will also look at the nuances required to build forecasting models such as: the impact of competitive forces on prices and costs, the effects of inflation and deflation, and long-term forecasting considerations.

2. Building a Financial Statement Model

Financial modeling typically begins with forecasting items on the income statement.

Income Statement Modeling: Revenue

Most companies have multiple sources of revenue. A company's revenue sources can be broken down by:

- Geographical regions
- Business segments
- Product lines

Once the sources of revenue are known, an analyst can use the following approaches to forecast revenue:

- The 'top-down' approach begins at the level of the overall economy. We then move down to sector, industry, and market for a specific product to forecast the revenue for an individual company.
- The 'bottom-up' approach begins at the level of the individual company or a unit within the company (e.g., a product line). We then sum up the projections for the individual products to forecast the total revenue for the company.
- The 'hybrid' approach combines top-down and bottom-up approaches. By using elements from both approaches, a hybrid approach can reveal implicit assumptions or errors that may arise from using a single approach.

Top-down Approaches to Modeling Revenue

Two common top-down approaches to modeling revenue are:

Growth relative to GDP growth approach: In this approach, we first forecast the growth rate of nominal GDP. We then forecast the company's revenue growth relative to GDP growth. For example, we can assume that the company's revenue will grow at a rate of 150 bps above the

nominal GDP growth rate.

The forecast may also be in relative terms. For example, if we forecast that the GDP will grow at 4%, and we believe that the company's revenue will grow at a 20% faster rate, then the forecasted increase in the company's revenue is $4\% \times (1 + 0.20) = 4.8\%$.

Market growth and market share: In this approach, we combine forecasts of growth in particular markets with forecasts of a company's market share. For example, assume Tesla is expected to maintain a market share of 1% in the automobile market. If the automobile market is expected to grow to \$30 billion in annual revenue, then Tesla's annual revenue is forecasted to grow to $1\% \times \$30 \text{ billion} = \300 million .

Bottom-Up Approaches to Modeling Revenue

Examples of bottom-up approaches include:

- *Time-series:* Forecasts based on historical growth rates or time-series analysis. For example, we may assume that the historical growth rate will continue, or that the growth rate will decline linearly from current rates to some long-run rate.
- *Return on capital:* Forecasts based on balance sheet accounts. For example, a bank's interest revenue can be calculated as loans multiplied by the average interest rate.
- *Capacity-based measure:* For example, a retailer's revenue may be forecasted based on same-store sales growth and sales related to new stores.

Hybrid Approaches to Modeling Revenue

Hybrid approaches are the most commonly used approaches in practice. They combine elements of both top-down and bottom-up approaches. For example, we may use a market growth and market share approach to model individual product lines or business segments (top-down), and then combine the individual projections to arrive at a forecast for the overall company (bottom-up).

Income Statement Modeling: Operating Costs

Operating costs include cost of goods sold (COGS) and selling, general, and administrative expenses (SG&A). Disclosures about operating costs are often less detailed than revenue. If information is available, then we can match cost analysis to revenue analysis. For example, costs may be modeled separately for different geographic regions, business segments or product lines.

Similar to revenue forecasting, costs can also be forecasted using a top-down, bottom-up, or hybrid approach:

- *Top-down approach:* Consider factors such as overall level of inflation, or industry-specific costs.
- *Bottom-up approach:* Consider factors such as segment-level margins, historical cost-

growth rates, historical margin levels, etc.

- *Hybrid view:* Incorporate elements from both top-down and bottom-up approaches.

Some points that analysts must consider when projecting operating costs are:

- Since variable costs are linked to revenue growth, they can be forecasted as a percentage of revenue.
- Since fixed costs are not directly related to revenue, they are assumed to grow at their own rate or at the rate of future PP&E growth.
- Determine whether a company has economies of scale at the current level of output. Economies of scale means the average costs per unit of a good produced falls as volume increases. Gross and operating margins tend to be positively related to sales, if there are economies of scale.
- Be aware of the uncertainty related to cost estimates such as reserve accounts, competitive factors and technological developments. For example, banks and insurance companies create reserves against estimated future losses. However, the actual losses may not be known for many years.

Cost of Goods Sold

Since COGS are directly related to sales, we can forecast future COGS as a percentage of future sales. Analysts should understand the historical relationship between COGS and sales, and determine if this relationship is likely to decrease, increase, or remain unchanged.

$\text{Sales} - \text{COGS} = \text{gross margin}$. Therefore, COGS and gross margin are inversely related.

Some factors that analysts should consider while forecasting COGS are:

- Forecasting accuracy can be improved by forecasting COGS for the company's various segments or product categories separately.
- Determine if a company has employed hedging strategies to protect its gross margins. When input prices increase, COGS increase and gross margin decreases. However, hedging strategies can help mitigate this impact.
- Examine gross profit margins of competitors. This can be a useful cross check for estimating a realistic gross margin. Any difference in gross margins for companies in the same segment must relate to differences in their business operations.

SG&A Expenses

As compared to COGS, SG&A expenses are less directly related to revenue.

SG&A can be broken down into two components – fixed and variable. The fixed components such as R&D expenses, management salaries, head office expenses, supporting IT and administrative operations tend to increase and decrease gradually over time. They do not

fluctuate with sales.

On the other hand, the variable components such as selling and distribution costs are more directly related to sales. For example, when sales increase, the company may pay out higher bonuses to its salesforce, it may also hire additional salespersons.

Income Statement Modelling: Non-Operating Costs and Other Items

Non-operating costs appear below operating profit on the income statement. The two significant non-operating costs that need to be forecasted are financing expenses (i.e. interest) and taxes.

Financing Expenses

Interest expense depends on the amount of debt on the balance sheet, as well as the interest rate associated with the debt. To forecast financing expenses, we first determine the capital structure of the company. The projected level of debt and the corresponding interest rates are then used to forecast the interest expense. To improve the quality of forecasts, we can use information provided in the notes to the financial statements about the maturity structure of the company's debt.

Some companies may also have interest income from investments. Interest income depends on the amount of cash and investments on the balance sheet, and the rates of return earned on investments. It is a key component of revenue for banks and insurance companies, but less important for non-financial companies.

Corporate Income Tax

There are three types of tax rates:

- *Statutory tax rate*: The tax rate established by law.
- *Effective tax rate*: Tax amount reported on the income-tax statement divided by the pre-tax income.
- *Cash tax rate*: Tax actually paid divided by pre-tax income.

Differences between cash taxes and reported taxes occur due to timing differences between accounting and tax calculations. These differences are reflected as a deferred tax asset or a deferred tax liability on the balance sheet.

Statutory tax rate and effective tax rates can be different for a number of reasons, such as tax credits, adjustments to previous years, withholding tax on dividends, etc.

The effective tax rate of a company that operates in multiple countries is the weighted average of the effective tax rates in each country. If the company is expected to report higher profits in a country with a high tax rate and lower profits in a country with a low tax rate, then its effective tax rate will increase. The effective tax rate is used to forecast earnings on the income statement and the cash tax rate is used to forecast cash flows.

To improve the quality of forecasts, the analyst should adjust for one-time events. Information provided in the notes to the financial statements can be used to identify such events.

If a company's effective tax rates are consistently lower than statutory rates or the effective tax rates reported by competitors, then analysts must scrutinize the reason when forecasting tax expense.

Income Statement Modeling: Other Items

A few other items on the income statement that can be forecasted are:

Dividends: Dividends are often forecasted to grow each year by a certain dollar amount or as a proportion of net income.

Income from affiliates: There are two possibilities here, depending on a company's ownership in an affiliate:

- More than 50%: consolidate the affiliate's results with its own. Report the portion of income that does not belong to the parent as minority interest.
- Less than 50%: Report its share of income from the affiliate under the equity method.

Share count: Share count can change under three circumstances: dilution related to stock options and convertible bonds, issuance of new shares, and share repurchases. Analysts must consider the market price of a stock and the capital structure of the company when projecting share count.

Unusual charges: These are difficult to predict and are usually not forecasted.

Balance Sheet and Cash Flow Statement Modeling

Income statement modeling is the foundation for modeling balance sheet and cash flow statements. Analysts can choose whether to focus on the balance sheet or the cash flow statement. Here we focus on the balance sheet. Cash flow statement can be derived based on the income statement and balance sheet.

Balance sheet modeling

Some balance sheet items flow directly from the forecasted income statement. For example, net income minus dividends will flow through to retained earnings.

Other balance sheet items are closely linked to the forecasted income statement and can be projected based on their historical relationship with income statement items. For example, working capital accounts are forecasted through efficiency ratios.

- Projected accounts receivable = forecasted sales x days sales outstanding/365
- Projected inventory = forecasted COGS / inventory turnover

If efficiency ratios are held constant, then working capital accounts will grow at the same

rate as income statement accounts.

Long-term assets such as net PP&E are less directly related to the income statement. PP&E depends on capital expenditures and depreciation. Depreciation forecasts are based on historical depreciation, whereas capital expenditure forecasts are based on the analyst's estimate of the future need for new PP&E. To improve the quality of forecasts, capital expenditure can be broken down into maintenance capital expenditure (needed to sustain current business) and growth capital expenditure (needed to expand the business). Each component can be forecasted separately. Due to inflation, maintenance capital expenditure forecasts should normally be higher than historical depreciation.

Finally, to forecast the capital structure, analysts use leverage ratios (such as debt to equity, debt to capital).

Summary

The general steps while building a financial statement model are summarized below.

1. Perform an industry overview using the Porter's five forces model.
 2. Perform a company overview to account for company specific factors. If applicable, divide the company's revenue sources into different segments.
 3. Construct a pro-forma income statement.
 - a. Revenue can be forecasted using a top-down, bottom-up or hybrid approach.
 - b. COGS can be forecasted as a percentage of sales using historical relationships.
 - c. In SG&A, selling and distribution costs such as wages are variable and can be estimated as a percentage of sales. General and administrative expenses are more or less fixed or increase/decrease gradually.
 - d. Financing costs can be forecasted using the expected level of debt and the corresponding interest rates.
 - e. Income tax expense can be forecasted based on the trends in the historical effective tax rate.
 4. Construct a pro-forma balance sheet.
 - a. Retained earnings will flow from the income statement.
 - b. Working capital accounts such as inventory, accounts receivables and accounts payable can be forecasted through efficiency ratios and corresponding income statement accounts.
 - c. PP&E can be forecasted based on projected capital expenditures (both maintenance and growth) and depreciation.
 5. Construct a pro-forma cash flow statement using the income statement and the balance sheet.
 6. Determine the intrinsic value of the company using an appropriate valuation model.
 - a. Forecast earnings or some other cashflow measure over the forecast horizon.
 - b. Determine a terminal value using a multiples-based approach or DFC approach.
 - c. Determine an appropriate discount rate for the company.
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3. Behavioral Finance and Analyst Forecasts

Five key biases that influence analyst forecasts are:

Bias	Description	Effect	Remedial action
Overconfidence Bias	A bias in which people demonstrate unwarranted faith in their own abilities.	Overconfidence arises more frequently when making contrarian predictions that others do not expect.	Analysts should record and share their forecasts and review them regularly, identifying both the correct and incorrect forecasts they have made.
Illusion of Control Bias	Tendency to overestimate the ability to control what cannot be controlled and to take ultimately fruitless actions in pursuit of control.	Analysts believe that forecasts can be made more accurate in two ways: by acquiring more information and opinions from experts and by creating more granular and complex models.	Restrict modeling variables to those that are regularly disclosed by the company, focus on the most important or impactful variables, and speak only with those who are likely to have unique or significant perspectives.
Conservatism Bias	People maintain their prior views or forecasts by inadequately incorporating new information.	Analysts do not update their forecasts after receiving conflicting information, such as disappointing earnings results or a competitor action.	Periodic reviews of forecasts and models by an investment team at a regular interval, such as each quarter. Create flexible models with fewer variables, to make changing assumptions easier

Representativeness Bias	Tendency to classify information based on past experiences and known classification.	Leads to base-rate neglect in which a phenomenon's rate of incidence in a larger population, or characteristics of a larger class to which a specific member belongs—its base rate—is neglected in favor of situation- or member-specific information.	Start with the base rate but determine which factors make the target company different from the base rate or class average and what the implications of those differences are, if any.
Confirmation Bias	Tendency to look for and notice what confirms prior beliefs and to ignore or undervalue whatever contradicts them.	Example: An analyst who has a positive view on a company might speak only to other analysts who share that view and the company's management, all of whom will likely tell the analyst what they want to hear and already know.	Speak to or read research from analysts with a negative opinion on the security under scrutiny. Seek perspectives from colleagues who are not economically or psychologically invested in the subject security.

4. The Impact of Competitive Factors in Prices and Costs

Competitive factors affect a company's ability to negotiate lower input prices with suppliers and to raise prices for products and services. Porter's five forces framework can be used as a basis for identifying such factors. The table below summarizes this framework.

Force	Comment
Threat of substitutes	Fewer substitutes or higher switching costs increase industry profitability.
Internal rivalry	Lower rivalry increases industry profitability.
Supplier power	Presence of many suppliers limits their pricing power, which in turn does not put a downward pressure on the industry's profitability.
Customer power	Presence of many buyers limits their negotiating power, which in turn does not put a downward pressure on the industry's profitability.

Threat of new entrants	High barriers to entry increase industry profitability.
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5. Modeling Inflation and Deflation

Inflation is an increase in the price of goods and services, whereas deflation is a decrease in the prices of goods and services. They can significantly affect the accuracy of forecasts of revenue, profit and cash flow. However, the degree of impact varies from company to company. Some companies are better able to pass on higher input costs to their customers by selling their output at higher prices. Such companies are more likely to have higher and more stable profits and cash flow, relative to competitors.

Sales Projections with Inflation and Deflation

We will now analyze how inflation and deflation affect industry and company sales forecasts.

Industry sales and inflation/deflation

- Higher input costs usually result in higher prices for end products. The industry structure determines how fast the increase in input prices is transmitted to output prices.
- Suppose input costs go up by 5%. To offset this, the selling price is increased by 5% and the same sales volume is maintained. Under this scenario, the gross profit margin percentage would be the same but the absolute amount of gross profit would increase.
- However, most products have an elastic demand. Therefore, when prices are increased, sales volume decreases.
- The decline in demand will not only depend on the elasticity of demand but also on the availability of substitutes, competitor behavior, etc. In an inflationary environment, companies that raise prices too soon will experience volume losses. Whereas companies that raise prices too late will experience declining profit margins.

Company sales and inflation/deflation

When forecasting the revenue of a company, identify a company's main input costs and the company's strategy for how it responds to rising input prices. This will help in determining the pricing. Will it pass on the increase in costs to customers, or accept margin reduction to retain market share? Revenue projections are based on expected price and volume development.

We consider the following factors to analyze the effect of inflation:

- *Price elasticity of the products:* If demand is relatively price inelastic, revenue will benefit from inflation. If demand is price elastic, revenues will decrease even if unit prices are increased because volumes decline.
- *Different rates of cost inflation:* Consider the geographic mix of a company's

operations to reflect different rates of the cost of inflation among countries. High inflation in export market might increase sales in foreign currency terms, but that gain might be lost if the foreign currency depreciates.

- *Inflation in costs of a company's individual product category:* For example, an increase in grain prices is likely to impact a bakery more than a supermarket.

Cost Projections with Inflation and Deflation

We will now analyze how inflation and deflation affect industry and company cost forecasts.

Industry costs and inflation/deflation

- Understanding the purchasing characteristics of an industry can be useful in forecasting costs. For example, if the industry uses hedging instruments such as long-term price-fixed forward contracts, then input price increases will be gradual instead of a sudden hike.
- Analysts should monitor the underlying drivers of input prices. For example, weather conditions can have a significant impact on the price of agricultural products and consequently on the input costs of industries that rely on agricultural products.
- The impact of inflation or deflation on an industry's cost structure depends on the competitive structure. For example, if companies have access to alternative inputs or are vertically integrated, the impact of volatility of input costs will be low.
- An analyst may also examine if a company can cut costs in other areas such as marketing and advertising to maintain margins in the short term.

Company costs and inflation/deflation

- While forecasting a company's costs, break down the cost structure by category and geography. For each cost element, assess the impact of inflation and deflation on input prices.
- Consider the company's ability to use cheaper alternatives for expensive inputs when input prices rise.

6. The Forecast Horizon and Long-Term Forecasting

Forecast time horizon is influenced by factors such as:

- Investment strategy for which the stock is being considered: For most investors, the forecast horizon is usually the expected holding period of the stock. Holding period corresponds to the portfolio turnover: average holding period = $1/\text{portfolio turnover}$. For example, an annual portfolio turnover of 20% indicates an average holding period of 5 years.
- Cyclicality of the industry: The forecast period should be long enough for the business

to reach the mid-cycle level of sales and profitability.

- Company specific factors: Factors such as recent acquisition or restructuring can also influence the forecast horizon. The forecast period should be long enough to allow the expected benefits from such an activity to be realized.
- Analyst's employer preferences: An analyst's employer may specify what forecast horizon to use.

Longer-term projections often provide a better representation of the normalized earnings potential of a company than a short-term forecast, especially when certain temporary factors are present. **Normalized earnings** are the expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that impact profitability (either positively or negatively).

"Growth relative to GDP growth" and "market growth and market share" methods can be applied to develop longer term revenue projections. Once revenue is projected, previously described methods are used to forecast costs and to complete the income statement, balance sheet and cash flow statement.

Along with earnings projections over the forecast period, analysts also determine the terminal value of the stock at the end of the forecast period. Terminal values are usually estimated using a multiples-based approach or a DCF approach.

When using a multiples-based approach, the choice of the multiple should be consistent with the long-run expectations for growth and required return. Generally, a historical multiple is used as a basis for the target multiple. If the future growth is likely to differ from the historical average, then the target multiple should include a premium or discount to account for this difference. For example, say a stock's median P/E over the last decade was 12. If the company is expected to perform very well in the future, then the P/E multiple should be adjusted upward while calculating the terminal value (and vice versa).

When using a DCF approach, two important parameters are the terminal year cash flow projections and the long-term growth rate. The terminal year cash flow should be normalized to a mid-cycle value to remove the impact of short-term events. An appropriate long-term perpetual growth rate should be determined that matches the future outlook of the company. Using an unrealistic long-term growth rate will make the terminal value projections incorrect. The greatest challenge that analysts face while using a DCF approach is anticipating inflection points, i.e., when the future will look significantly different from the past. Inflection points can occur due to: business cycles, economic disruptions, regulatory and technological changes, etc.

Summary

LO: Demonstrate the development of a sales-based pro forma company model

The general steps while building a financial statement model are summarized below.

1. Perform an industry overview using the Porter's five forces model.
2. Perform a company overview to account for company specific factors. If applicable, divide the company's revenue sources into different segments.
3. Construct a pro-forma income statement.
 - a. Revenue can be forecasted using a top-down, bottom-up or hybrid approach.
 - b. COGS can be forecasted as a percentage of sales using historical relationships.
 - c. In SG&A, selling and distribution costs such as wages are variable and can be estimated as a percentage of sales. General and administrative expenses are more or less fixed or increase/decrease gradually.
 - d. Financing costs can be forecasted using the expected level of debt and the corresponding interest rates.
 - e. Income tax expense can be forecasted based on the trends in the historical effective tax rate.
4. Construct a pro-forma balance sheet.
 - a. Retained earnings will flow from the income statement.
 - b. Working capital accounts such as inventory, accounts receivables and accounts payable can be forecasted through efficiency ratios and corresponding income statement accounts.
 - c. PP&E can be forecasted based on projected capital expenditures (both maintenance and growth) and depreciation.
5. Construct a pro-forma cash flow statement using the income statement and the balance sheet.
6. Determine the intrinsic value of the company using an appropriate valuation model.
 - a. Forecast earnings or some other cashflow measure over the forecast horizon.
 - b. Determine a terminal value using a multiples-based approach or DFC approach.
 - c. Determine an appropriate discount rate for the company.

LO: Explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases.

Five key behavioral biases that influence analyst forecasts are overconfidence, illusion of control, conservatism, representativeness, and confirmation bias.

LO: Explain how the competitive position of a company based on a Porter's five forces analysis affects prices and costs.

Competitive factors affect a company's ability to negotiate lower input prices with suppliers and to raise prices for products and services. Porter's five forces framework can be used as a basis for identifying such factors.

Force	Comment
Threat of substitutes	Fewer substitutes or higher switching costs increase industry profitability.
Internal rivalry	Lower rivalry increases industry profitability.
Supplier power	Presence of many suppliers limits their pricing power, which in turn does not put a downward pressure on the industry's profitability.
Customer power	Presence of many buyers limits their negotiating power, which in turn does not put a downward pressure on the industry's profitability.
Threat of new entrants	High barriers to entry increase industry profitability.

LO: Explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation.

Effect on sales:

- Higher input costs usually result in higher prices for end products. Since most products have an elastic demand, when prices are increased sales volume decrease.
- In an inflationary environment, companies that raise prices too soon will experience volume losses. Whereas, companies that raise prices too late will experience declining profit margins.

Effect on costs:

- If the company uses hedging instruments such as long-term forward contracts, then input price increases will be gradual instead of a sudden hike.
- If companies have access to alternative inputs or are vertically integrated, the impact of volatility of input costs will be low.

LO: Explain considerations in the choice of an explicit forecast horizon and an analyst's choices in developing projections beyond the short-term forecast horizon.

The factors to consider when choosing an explicit time horizon include the projected holding period, an investor's average portfolio turnover, cyclicalities of an industry, company specific factors, and employer preferences.

Along with earnings projections over the forecast period, analysts also determine the terminal value of the stock at the end of the forecast period. Terminal values are usually estimated using a multiples-based approach or a DCF approach.

- While using the multiples-based approach, if the future growth is likely to differ from the historical average, then the target multiple should include a premium or discount

to account for this difference.

- While using the DCF approach, the terminal year cash flow should be normalized to a mid-cycle value, to remove the impact of short-term events. An appropriate long-term perpetual growth rate should be determined that matches the future outlook of the company.